

Report

Autor: Artur Mizuła

Test algorytmów powiększania

IMG_SMALL/SMALL_0001.tif

Original



NN scale 200 Bilinear scale 200



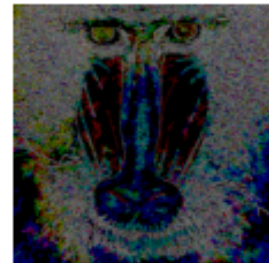
Edges Original



Edges NN



Edges Bilinear



IMG_SMALL/SMALL_0001.tif

Original



NN scale 300 Bilinear scale 300



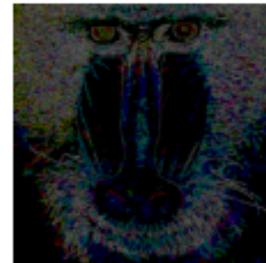
Edges Original



Edges NN



Edges Bilinear



IMG_SMALL/SMALL_0001.tif

Original



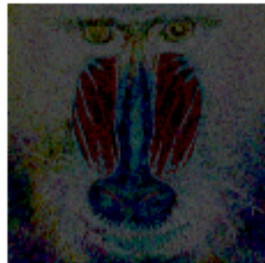
NN scale 500 Bilinear scale 500



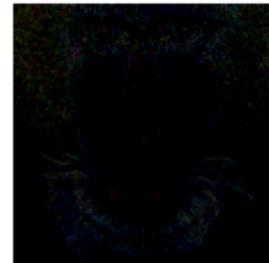
Edges Original



Edges NN



Edges Bilinear



IMG_SMALL/SMALL_0002.png

Original



NN scale 200 Bilinear scale 200



Edges Original



Edges NN



Edges Bilinear



IMG_SMALL/SMALL_0002.png

Original



NN scale 300 Bilinear scale 300



Edges Original



Edges NN



Edges Bilinear



IMG_SMALL/SMALL_0002.png

Original



NN scale 500 Bilinear scale 500



Edges Original



Edges NN



Edges Bilinear



IMG_SMALL/SMALL_0003.png

Original



NN scale 200 Bilinear scale 200



Edges Original



Edges NN



Edges Bilinear



IMG_SMALL/SMALL_0003.png

Original



NN scale 300 Bilinear scale 300



Edges Original



Edges NN



Edges Bilinear



IMG_SMALL/SMALL_0003.png

Original



NN scale 500 Bilinear scale 500



Edges Original



Edges NN



Edges Bilinear



IMG_SMALL/SMALL_0004.jpg

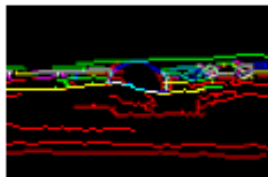
Original



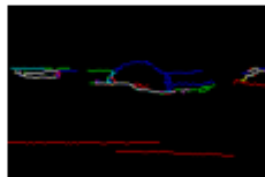
NN scale 200 Bilinear scale 200



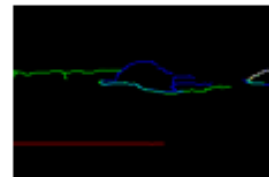
Edges Original



Edges NN



Edges Bilinear



IMG_SMALL/SMALL_0004.jpg

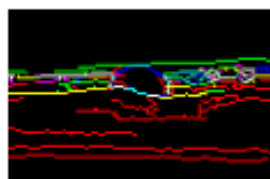
Original



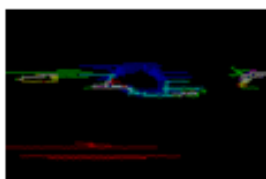
NN scale 300 Bilinear scale 300



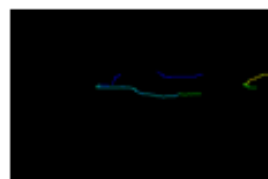
Edges Original



Edges NN



Edges Bilinear



IMG_SMALL/SMALL_0004.jpg

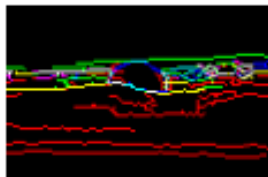
Original



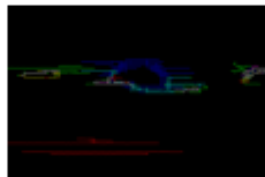
NN scale 500 Bilinear scale 500



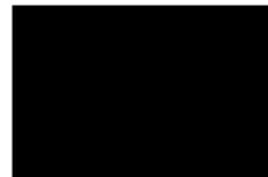
Edges Original



Edges NN



Edges Bilinear



IMG_SMALL/SMALL_0005.jpg

Original



NN scale 200 Bilinear scale 200



Edges Original



Edges NN



Edges Bilinear



IMG_SMALL/SMALL_0005.jpg

Original



NN scale 300 Bilinear scale 300



Edges Original



Edges NN



Edges Bilinear



IMG_SMALL/SMALL_0005.jpg

Original



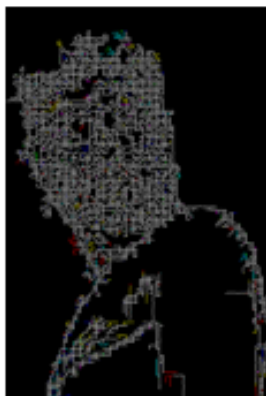
NN scale 500 Bilinear scale 500



Edges Original



Edges NN



Edges Bilinear



IMG_SMALL/SMALL_0006.jpg

Original



NN scale 200 Bilinear scale 200



Edges Original



Edges NN



Edges Bilinear



IMG_SMALL/SMALL_0006.jpg

Original



NN scale 300 Bilinear scale 300



Edges Original



Edges NN



Edges Bilinear



IMG_SMALL/SMALL_0006.jpg

Original



NN scale 500 Bilinear scale 500



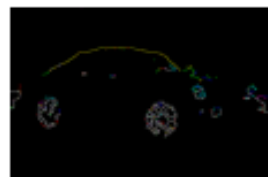
Edges Original



Edges NN



Edges Bilinear



IMG_SMALL/SMALL_0007.jpg

Original



NN scale 200 Bilinear scale 200



Edges Original



Edges NN



Edges Bilinear



IMG_SMALL/SMALL_0007.jpg

Original



NN scale 300 Bilinear scale 300



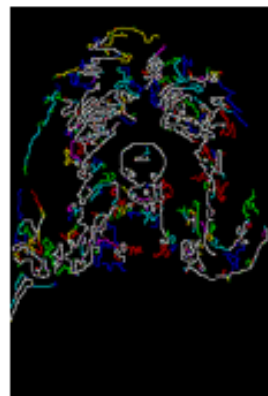
Edges Original



Edges NN



Edges Bilinear



IMG_SMALL/SMALL_0007.jpg

Original



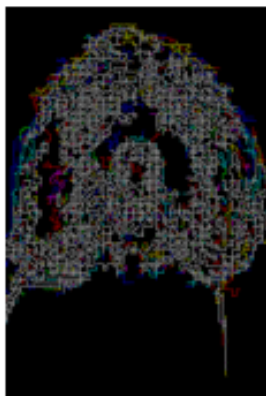
NN scale 500 Bilinear scale 500



Edges Original



Edges NN



Edges Bilinear



IMG_SMALL/SMALL_0008.jpg

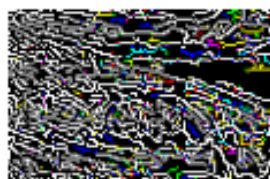
Original



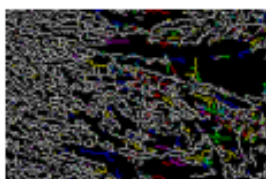
NN scale 200 Bilinear scale 200



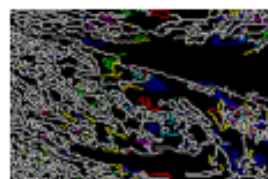
Edges Original



Edges NN



Edges Bilinear



IMG_SMALL/SMALL_0008.jpg

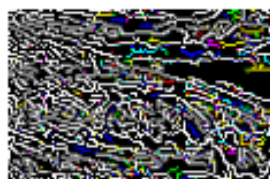
Original



NN scale 300 Bilinear scale 300



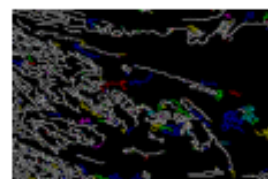
Edges Original



Edges NN



Edges Bilinear



IMG_SMALL/SMALL_0008.jpg

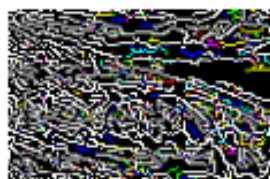
Original



NN scale 500 Bilinear scale 500



Edges Original



Edges NN



Edges Bilinear



IMG_SMALL/SMALL_0009.jpg

Original



NN scale 200 Bilinear scale 200



Edges Original



Edges NN



Edges Bilinear



IMG_SMALL/SMALL_0009.jpg

Original



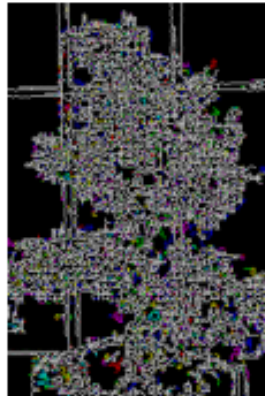
NN scale 300 Bilinear scale 300



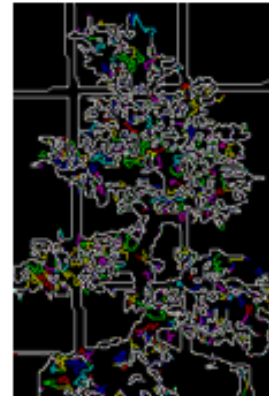
Edges Original



Edges NN



Edges Bilinear



IMG_SMALL/SMALL_0009.jpg

Original



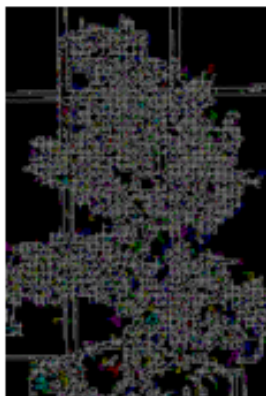
NN scale 500 Bilinear scale 500



Edges Original



Edges NN



Edges Bilinear



IMG_SMALL/SMALL_0010.jpg

Original



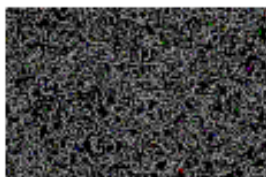
NN scale 200 Bilinear scale 200



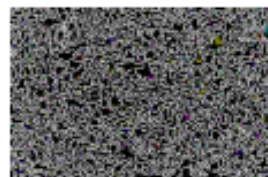
Edges Original



Edges NN



Edges Bilinear



IMG_SMALL/SMALL_0010.jpg

Original



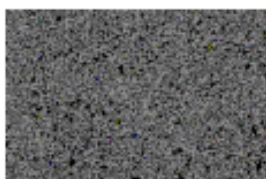
NN scale 300 Bilinear scale 300



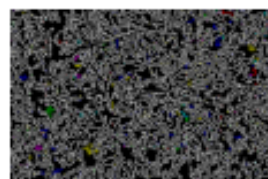
Edges Original



Edges NN



Edges Bilinear



IMG_SMALL/SMALL_0010.jpg

Original



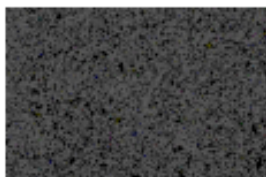
NN scale 500 Bilinear scale 500



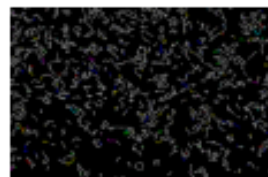
Edges Original



Edges NN



Edges Bilinear



Test algorytmów pomniejszania

IMG_BIG/BIG_0001.jpg ROI: [1800, 1500, 1200, 1200]

Original



NN scale 5



Bilinear scale 5



Edges Original



Edges NN



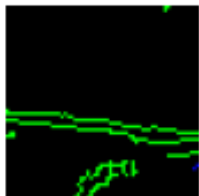
Edges Bilinear



Mean Resizing scale 5

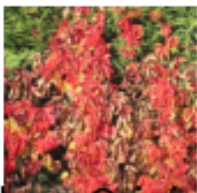


Edges Mean Resizing



IMG_BIG/BIG_0001.jpg ROI: [1800, 1500, 1200, 1200]

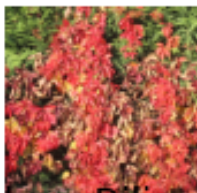
Original



NN scale 25



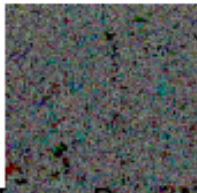
Bilinear scale 25



Edges Original



Edges NN



Edges Bilinear



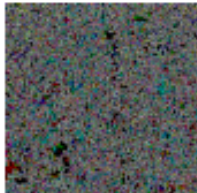
Mean Resizing scale 25



Edges Mean Resizing



Edges Weighted Mean Resizing

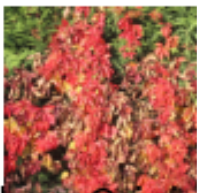


Edges Median Resizing



IMG_BIG/BIG_0001.jpg ROI: [1800, 1500, 1200, 1200]

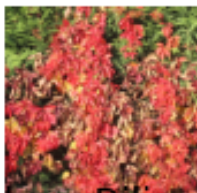
Original



NN scale 50



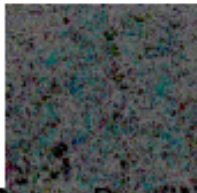
Bilinear scale 50



Edges Original



Edges NN



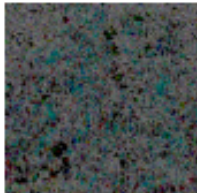
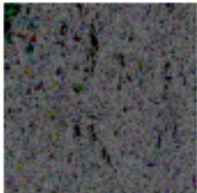
Edges Bilinear



Mean Resizing scale 50 Weighted Mean Resizing scale 50 Median Resizing scale 50

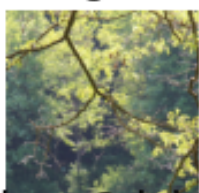


Edges Mean Resizing Edges Weighted Mean Resizing Edges Median Resizing



IMG_BIG/BIG_0002.jpg ROI: [2200, 300, 1200, 1200]

Original



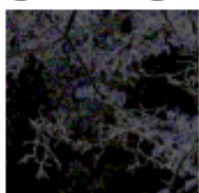
NN scale 5



Bilinear scale 5



Edges Original



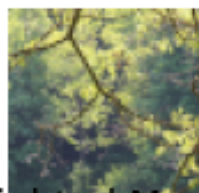
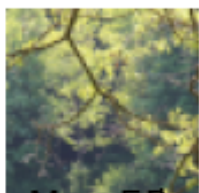
Edges NN



Edges Bilinear



Mean Resizing scale 5

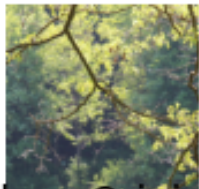


Edges Mean Resizing

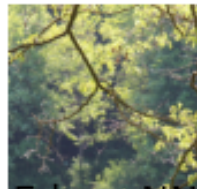


IMG_BIG/BIG_0002.jpg ROI: [2200, 300, 1200, 1200]

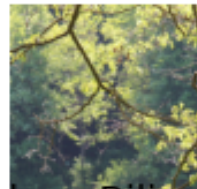
Original



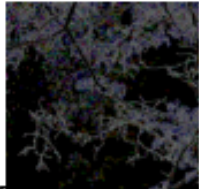
NN scale 25



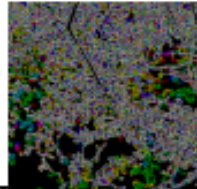
Bilinear scale 25



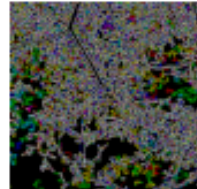
Edges Original



Edges NN



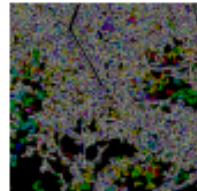
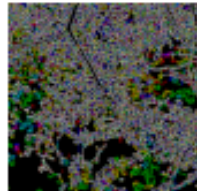
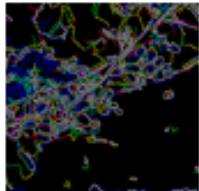
Edges Bilinear



Mean Resizing scale 25



Edges Mean Resizing



IMG_BIG/BIG_0002.jpg ROI: [2200, 300, 1200, 1200]

Original



NN scale 50



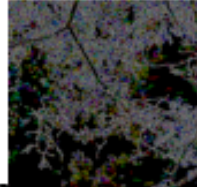
Bilinear scale 50



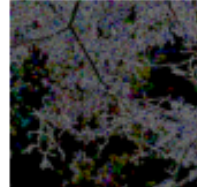
Edges Original



Edges NN



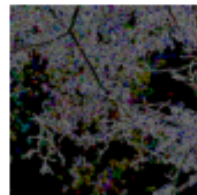
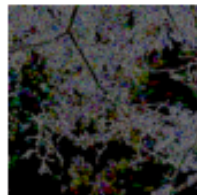
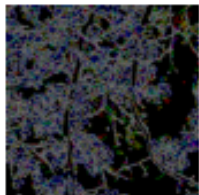
Edges Bilinear



Mean Resizing scale 50



Edges Mean Resizing



IMG_BIG/BIG_0003.jpg ROI: [450, 2550, 1200, 1200]

Original



NN scale 5



Bilinear scale 5



Edges Original



Edges NN



Edges Bilinear



Mean Resizing scale 5



Edges Mean Resizing



IMG_BIG/BIG_0003.jpg ROI: [450, 2550, 1200, 1200]

Original



NN scale 25



Bilinear scale 25



Edges Original



Edges NN



Edges Bilinear



Mean Resizing scale 25



Edges Mean Resizing



IMG_BIG/BIG_0003.jpg ROI: [450, 2550, 1200, 1200]

Original



NN scale 50



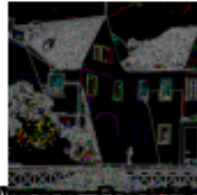
Bilinear scale 50



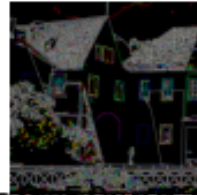
Edges Original



Edges NN



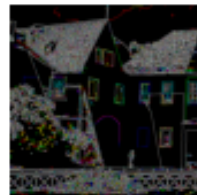
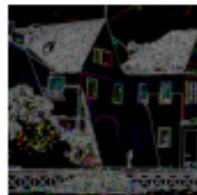
Edges Bilinear



Mean Resizing scale 50



Edges Mean Resizing

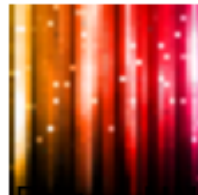


IMG_BIG/BIG_0004.png ROI: [600, 200, 700, 700]

Original



NN scale 5



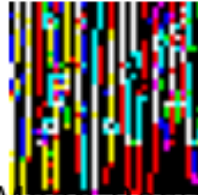
Bilinear scale 5



Edges Original



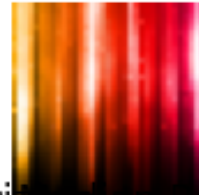
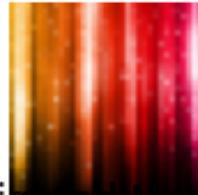
Edges NN



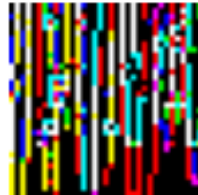
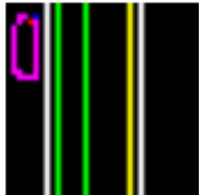
Edges Bilinear



Mean Resizing scale 5



Edges Mean Resizing



IMG_BIG/BIG_0004.png ROI: [600, 200, 700, 700]

Original

NN scale 25

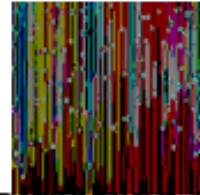
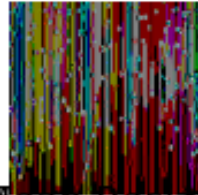
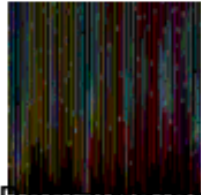
Bilinear scale 25



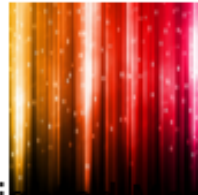
Edges Original

Edges NN

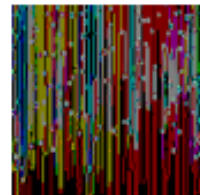
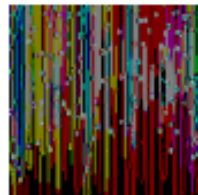
Edges Bilinear



Mean Resizing scale 25



Edges Mean Resizing



IMG_BIG/BIG_0004.png ROI: [600, 200, 700, 700]

Original

NN scale 50

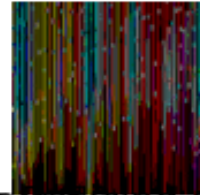
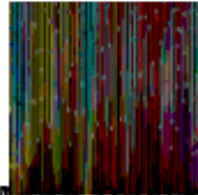
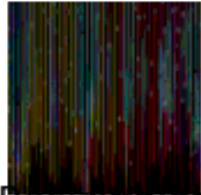
Bilinear scale 50



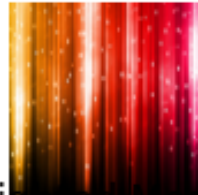
Edges Original

Edges NN

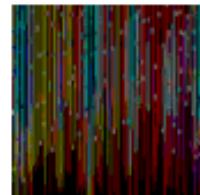
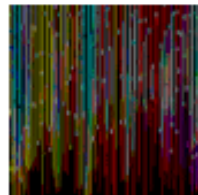
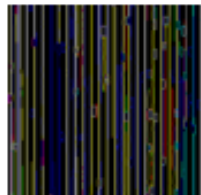
Edges Bilinear



Mean Resizing scale 50



Edges Mean Resizing



Podsumowanie i wnioski

Metoda najbliższego sąsiada i interpolacja biliniowa poprawnie wizualizują obraz przy powiększaniu. Przy większym powiększaniu występuje wygładzanie krawędzi. Metoda NN lepiej zachowuje ostrość

Jeśli chodzi o testowanie algorytmów pomniejszania przy silnym pomniejszaniu obrazu Metoda najbliższego sąsiada i biliniowa tworzą pikselowy efekt. Reszta algorytmów radzi sobie lepiej wygładzając obraz co powoduje brak pikselowego efektu